

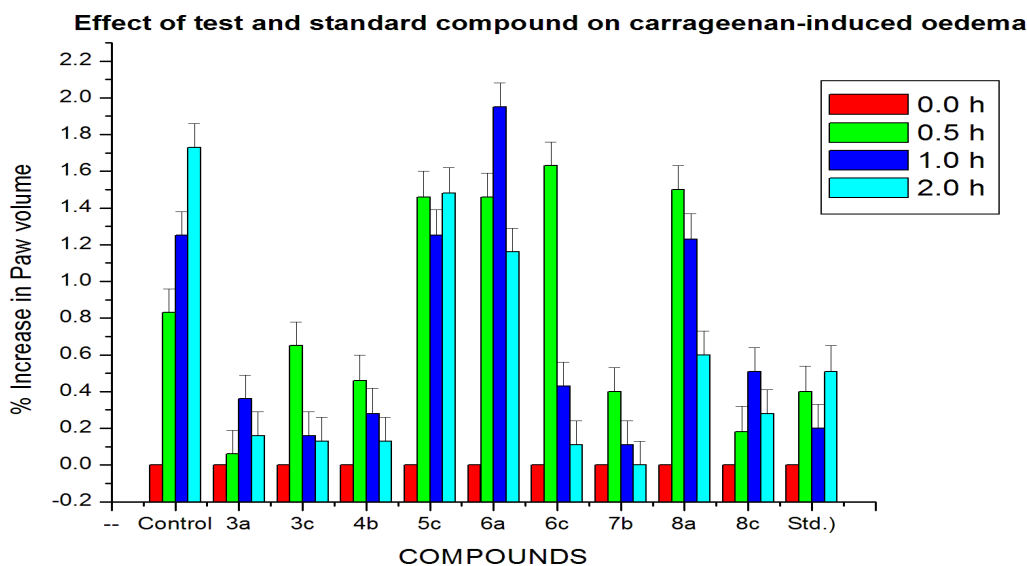
Anti-inflammatory activity

Table 5 effect of test and standard compound on carrageenan-induced oedema

Comp.	Dose (mg/kg,p.o)	Increase in paw volume (Mean±SEM)in ml at			
		0.0 h	0.5 h	1.0 h	2.0 h
Control(N/saline)	2.5/kg	0.0	0.83±0.13	1.25±0.13	1.73±0.13
3a	50mg/kg	0.0	0.06±0.13**	0.36±0.13**	0.16±0.13**
3c	50mg/kg	0.0	0.65±0.13*	0.16±0.13**	0.13±0.13**
4b	50mg/kg	0.0	0.46±0.14**	0.28±0.14**	0.13±0.13**
5c	50mg/kg	0.0	1.46±0.14	1.25±0.14	1.48±0.14
6a	50mg/kg	0.0	1.46±0.14	1.95±0.13	1.16±0.13
6c	50mg/kg	0.0	1.63±0.13	0.43±0.13	0.11±0.13
7b	50mg/kg	0.0	0.4±0.13**	0.11±0.13**	0.0±0.13**
8a	50mg/kg	0.0	1.5±0.13	1.23±0.14	0.6±0.13
8c	50mg/kg	0.0	0.18±0.14**	0.51±0.13**	0.28±0.13*
Std.(diclofenac)	50mg/kg	0.0	0.4±0.14**	0.2±0.13**	0.51±0.14**

The statistical significance of differences assessed with an analysis of variance (ANOVA), followed by Bonferroni t-test. Results are expressed in mean +_SEM. (N=6) significance levels *P<0.05 and **P<0.01, compared with control respectively.

Graph 3 comparison results are expressed in mean ± SEM for statistical significance of differences of above compounds.



RESULT AND DISCUSSION

Antimicrobial activity.

Derivatives of 1-methyl-4(5-(4-(phenyl amino) phenyl)-1,3,4-thiadiazol-2-e thiosemicarbazide (2a-f) were reacted with corresponding reactant dimethyl lamina, morpholine and piperidine (a-c) in the presence of formaldehyde and concentrated acid. All synthesized compounds were obtained in the range of 57.41-66.7% yield. Purity and homogeneity of the synthesized compound were routinely checked by sharp melting point and TLC. Melting point of the all synthesized compound was observed between at 240.0-251.0-°C.

57.41-66.7 % yield purity and homogeneity of the synthesized compound were routinely checked by sharp melting point and TLC. Melting point of the all synthesized compounds was observed 240.0 to 251.0-°C.

The *in vitro* antimicrobial activity of the synthesized compound was evaluated by cup plate method for bacteria viz. *S. aureus*, *B. subtilis*, *E. coli*, *K. pneumonia* and fungi *C. albicans*, *A. Niger*. Some compounds exhibited potent pharmacological activities as compared to the standard drugs.

The compounds 3b and 4a exhibited highest activity i.e. each with 82.60% inhibition against *S. aureus* while the compound 4b, 5b, 6b, 7c showed 65.21, 69.56, 52.17 and 52.17% inhibition respectively against *S. aureus*. Against *B. subtilis* the compound 3a, 3b, and 4c showed 85.71, 95.23, 66.66% inhibition respectively and were the highest activity compound. While the compound for 4a, 5a, 6c, 8b (each with 57.41% inhibition) and 5b (47.61% inhibition) showed activity.

When tested against *E. coli* the compound 3c, 4a, 6b, 7a and 7c exhibited 60.86, 52.17, 52.17, 52.17 and 60.86% inhibition respectively and these compounds showed good activity among the tested compound. The highest activity was observed with 8c (with 69.56% inhibition).

Against *K. pneumoniae* the compound 3b (65.21%), 4c (60.86% inhibition), 5a (60.86% inhibition), 7a (65.21% inhibition), 7b (65.21% inhibition) and 7c (65.21% inhibition) exhibited good activity. Among the highest activity tested compounds 3a (69.56%) showed activity.

All the synthesized compounds were screened *in vitro* against the pathogenic fungal strains *A. niger* and *C. albicans*. The compound 3a (68.42% inhibition), 5a (73.68% inhibition), 5c (63.15% inhibition), 6b, 7b (each with 63.15% inhibition) and 8a (68.42% inhibition) showed excellent activities.